

Practical 7 - Competing risks 2

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `aaatrial.dta`

Introduction

In this practical we will use the `aaatrial` data, as in Practical 6. This is a dataset from a randomised trial of screening for abdominal aortic aneurysm (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The key variables are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>group</code>	Randomisation group: 0=Control, 1=Invited
<code>dateran</code>	Date of randomisation
<code>aaadeath</code>	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
<code>allddeath</code>	All-cause mortality: 0=Censored, 1=Death (any cause)

Variable	Description
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit
dob	Date of birth

Aims

The aims of this session are:

- Fit and interpret results from cause-specific hazard models.
- Fit and interpret results from subdistribution hazard models.
- Estimate cumulative incidence functions based on cause-specific hazard models and based on a subdistribution hazard model.

Question 1

- Load the data.
- Fit cause-specific Cox models for AAA-death and non-AAA-death with `group` as the explanatory variable. For AAA-deaths this can be done using the code below.

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)
stcox i.group
```

- Write down the form of the hazard models and interpret the cause-specific hazard ratios.
- How is the competing event treated in the fitting of the cause-specific Cox models?

SOLUTION

- The form of the cause-specific hazard model for AAA-death (`deathtype=2`) is $h_2(t|X) = h_{0,2}(t)e^{\beta_2 X}$ and that for non-AAA-death (`deathtype=1`) is $h_1(t|X) = h_{0,1}(t)e^{\beta_1 X}$, where

$$h_k(t|X) = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \Pr(t \leq T < t + \delta t, D = k | T \geq t), k = 1, 2$$

and where $h_{0,k}(t)$ denotes the cause-specific hazard for event type k ($k = 1, 2$) at time t when $X = 0$.

- For AAA-death, the cause-specific HR for group is 0.67 (95% CI 0.45 to 0.99). This means that at any given time since randomisation, the hazard of AAA-death among individuals invited to screening and who have not yet died from another cause is 0.67 times the hazard for individuals in the control group. The 95% CI excludes the null value of 1, so there is evidence against the null hypothesis that this cause-specific hazard ratio is equal to 1.

- For non-AAA-death, the cause-specific HR for group is 0.98 (95% CI 0.92 to 1.04). This means that at any given time since randomisation, the hazard of non-AAA-death among individuals invited to screening who have not yet died from AAA-death is 0.98 times that among individuals in the control group. The 95% CI includes the null value of 1, so there is no evidence against the null hypothesis that this cause-specific hazard ratio is equal to 1.
- The cause-specific Cox hazard model for AAA-death is fitted using a partial likelihood in which individuals are excluded from the risk set for AAA-death after they have been censored or after they have died from another cause.
- The cause-specific Cox hazard model for non-AAA-death is fitted using a partial likelihood in which individuals are excluded from the risk set for non-AAA-death after they have been censored or after they have had an AAA-death.
- In the fitting of cause-specific hazard models individuals who die from the other cause are treated in the same way as people who are censored. This is fine for estimation of cause-specific hazard ratios.

Question 2

- In this question we will estimate the cumulative incidences of AAA-death and non-AAA-death based on the cause-specific hazard models fitted in question 1. This can be done in Stata using `stcompadj` as follows:

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)

stcompadj group=0, compet(1) maineffect(group) ///
    competeffect(group) gen(cinc_AAA_0 cinc_nonAAA_0)

stcompadj group=1, compet(1) maineffect(group) ///
    competeffect(group) gen(cinc_AAA_1 cinc_nonAAA_1)
```

- Take a look at the variables that have been added to the data set.
- What are the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the two treatments.
- We can plot the estimated cumulative incidences based on the cause-specific hazard models as follows (shown for AAA-death):

```
twoway (scatter cinc_AAA_0 _t, c(1) sort msym(p) ) ///
    (scatter cinc_AAA_1 _t, c(1) sort msym(p) ) , ///
    legend(lab(1 "Control, AAA death") lab(2 "Invited, AAA death"))
```

SOLUTION

- The `stcompadj` code gives us new variables in the data called `cinc_AAA_0` (cumulative incidence for AAA-death in group 0), `cinc_nonAAA_0` (cumulative incidence for non-AAA-death in group 0), `cinc_AAA_1` (cumulative incidence for AAA-death in group 1), `cinc_nonAAA_1` (cumulative incidence for non-AAA-death in group 1). These are listed at each event time in the data - the variables do NOT give us the cumulative incidence for the person in that row, but for the time (`_t`) in that row.
- For group 0 (control), the estimated cumulative incidence of non-AAA-death up to 5 years is 14% and the estimated cumulative incidence of AAA-death up to 5 years is 0.5%.
- For group 1 (invited), the estimated cumulative incidence of non-AAA-death up to 5 years is 13% and the estimated cumulative incidence of AAA-death up to 5 years is 0.3%.
- Note that these estimates are slightly different from those found in R - Stat and R appear to use slightly different ways of going from the cause-specific hazard models to the cumulative incidences.

Question 3

- Create a new variable that is the participant's age in years at the time of randomisation, using the date of birth (`dob`) and date of randomisation (`dateran`).
- Fit cause-specific Cox models for AAA-death and non-AAA-death with group and age as the explanatory variables.
- Write down the forms of the hazard models and interpret the cause-specific hazard ratios.

SOLUTION

- The form of the cause-specific hazard model for AAA-death (`deathtype=2`) is $h_2(t|X) = h_{0,2}(t)e^{\beta_2 grade + \gamma_2 age}$ and that for non-AAA-death (`deathtype=1`) is $h_1(t|X) = h_{0,1}(t)e^{\beta_1 grade + \gamma_1 age}$. Here $h_{0,k}(t)$ is the baseline cause-specific hazard at time t for event type k ($k = 1, 2$), referring to a person with `group=0` and `age=0`. The parameter β_k is the cause-specific log hazard ratio for group, and γ_k is the cause-specific log hazard ratio for a 1 year increase in age ($k = 1, 2$).
- AAA-death: The estimated cause-specific HR for group is 0.67 (95% CI 0.45-0.99), and that for age is 1.16 (95% CI 1.09-1.23). The interpretation is that the invited group (`group=1`) are estimated to have a hazard for AAA-death that is 33% lower than that for the uninvited group, conditional on age. The confidence interval excludes 1, so there is evidence of an effect of the intervention of AAA-death hazard, given age. Conditional on group, a 1-year increase in age is associated with a 16% increase in the hazard of AAA-death, and this is highly statistically significant.
- non-AAA-death: The estimated cause-specific HR for group is 0.98 (95% CI 0.92-1.04), and that for age is 1.10 (95% CI 1.09-1.11). The interpretation is that the invited group (`group=1`) is estimated to have a hazard for AAA-death that is 2% lower than that for

the uninvited group, conditional on age, but the confidence interval excludes 1, so there is no evidence of an effect of the intervention on the hazard of AAA-death, given age. Conditional on group, a 1-year increase in age is associated with a 10% increase in the hazard of AAA-death, and this is highly statistically significant.

Question 4

Based on the cause-specific hazard models fitted in question 3, obtain the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the following four individuals:

Person	Group	Age
1	0	65
2	1	65
3	0	70
4	1	70

You can do this by modifying the code from Question 2 (or by consulting the do file).

SOLUTION

- The cumulative incidence estimates at time 5 for the four people are shown below. Note that these estimates are slightly different from those found in R.

Person	Group	Age	Cumulative incidence at time 5: AAA-death	Cumulative incidence at time 5: non-AAA-death
1	0	65	0.3%	9%
2	1	65	0.2%	9%
3	0	70	0.5%	14%
4	1	70	0.4%	14%

Question 5

- In this question we will fit Fine and Gray models (i.e. subdistribution hazard models) for AAA-death, using group as the explanatory variable. This can be done using the `stcrreg` function in Stata as follows:

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)
stcrreg i.group, compete(deathtype=1)
```

- Interpret the subdistribution hazard ratios, and think about how the interpretation differs from that of the cause-specific hazard ratios.

SOLUTION

- The subdistribution HR estimate for group is 0.67 (95% CI 0.45-0.99). The interpretation here is that the group invited to screening have a 33% reduction in hazard of AAA death compared to the control group, *among those at a given time who have not yet died of AAA but who may have died of non-AAA death*. This last part of the previous sentence is the big difference of interpretation compared to the cause-specific hazard model.
- In this example the subdistribution HR for group is very similar to the cause-specific HR. This will be because the incidence of the competing event (non-AAA death) is similar in both groups. If the competing risks differed substantially between groups, we would see a bigger difference between the two types of hazard ratio.

Question 6

- Based on the Fine and Gray model from Question 5, obtain estimates of the cumulative incidences of AAA-death in the two groups using the code below.
- What is the estimated cumulative incidence up to 5 years.

```
** Baseline cumulative incidence (group=0)
predict cinc_fg_0, basecif

** Use this to calculate the cumulative incidence in the invited group
gen cinc_fg_1 = (1-(1-cinc_fg_0)^(exp(_b[1.group])))
```

SOLUTION

The estimated cumulative incidence of AAA-death up to 5 years is 0.5% in the control group and 0.3% in the invited group.

Question 7

Using a plot, compare the cumulative incidences of non-AAA-death in the two treatment groups based on three estimation methods: (i) the non-parametric Aalen-Johansen approach, (ii) cause-specific hazard models, (iii) a subdistribution (Fine and Gray) model.

Have a go at making the plot yourself, or see the example code in the solutions.

Explain why you may get different estimates using the three approaches.

SOLUTION

- In this case the cumulative incidence estimates from the three estimation methods are very similar - in Stata the cause-specific and Fine and Gray estimates appear to be identical in this example, but this will not be true in general.
- The non-parametric (Aalen-Johansen) approach does not rely on any modelling assumptions, i.e. it does not assume proportionality of hazards or any particular shape for the hazard.
- The cause-specific hazards approach assumes proportionality of cause-specific hazards between the two groups in the cause-specific models for both AAA-death and non-AAA-death, but does not assume any particular shape for the baseline hazard.
- The Fine and Gray approach assumes proportionality of subdistribution hazards for AAA-death between the two groups, but does not assume any particular shape for the baseline subdistribution hazard.